



极限的计算技巧

计算极限的值

一. 小结论

$$1. \lim_{n \rightarrow \infty} \sqrt[n]{a} = 1 \ (a > 0), \lim_{n \rightarrow \infty} \sqrt[n]{n} = 1, \lim_{n \rightarrow \infty} |q|^n = 0 \ (|q| < 1)$$

2. 无穷小量 $\times$ 有界变量 = 无穷小量

3. 无穷大量与无穷小量互为倒数关系

4. 极限运算形象推广:

$$\text{计算 } \lim_{x \rightarrow a} f(x) = ? \Rightarrow x \approx a \rightarrow f(x) \approx f(a)$$

二. 极限运算的一般想法

1. 无穷大的运算转化为有限数的运算 ($\frac{1}{\infty} = 0$)

2. 根号相减 $\xrightarrow[\text{有理化}]{\text{分子/分母}}$ 相加.

3. 数的无穷大次序考虑(正/负)无穷大(单侧极限)

4. 无穷项 $\rightarrow$ 有限项

$$5. \lim_{x \rightarrow a} f(x) = \frac{0}{0} \begin{cases} \text{分子分母约去因子 } (x-a) \\ \text{无穷小代换 } (x-a) \\ \text{洛必达/泰勒} \end{cases}$$

1. 无穷大 $\rightarrow$ 有限数

▲: 提取最大项, 无穷变有限.

$$\lim_{x \rightarrow \infty} \frac{(2x-1)^3(3-x)^2}{(3x-5)(x-2)^3(x+1)}$$

$$\lim_{x \rightarrow \infty} \frac{(2x-1)^3(3x+2)^{97}}{(3x+1)^{100}}$$

$$= \lim_{x \rightarrow \infty} \left(\frac{2x-1}{x-2} \right)^3 \lim_{x \rightarrow \infty} \frac{x^3(x-3)^2}{(x-1)(3x-5)} = \lim_{x \rightarrow \infty} \left(\frac{2x-1}{3x+1} \right)^3 \lim_{x \rightarrow \infty} \left(\frac{3x+2}{3x+1} \right)^{97}$$

$$= \frac{8}{3}$$

$$= \left(\frac{2}{3} \right)^3$$

$$= \frac{8}{27}$$

2. 根号

▲: 分子/分母有理化, 约分化简.

$$\begin{aligned} \textcircled{1} \lim_{x \rightarrow \infty} x(\sqrt{x^2+1} - x) &= \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2+1} + x} \\ &= \lim_{x \rightarrow \infty} \frac{1}{1 + \sqrt{1 + \frac{1}{x^2}}} \\ &= \frac{1}{2} \end{aligned}$$

分母有理化

$$\begin{aligned} \textcircled{2} \lim_{n \rightarrow \infty} \sin(\sqrt{n^2+1} \pi) \\ &= \lim_{n \rightarrow \infty} \sin(\sqrt{n^2+1} \pi - n\pi) (-1)^n \\ &= \lim_{n \rightarrow \infty} (-1)^n \sin \pi (\sqrt{n^2+1} - n) \\ &= 0 \end{aligned}$$

配凑 + 分母有理化

$$\begin{aligned} \lim_{n \rightarrow \infty} \sin^2(\sqrt{n^2+n} \pi) \\ &= \lim_{n \rightarrow \infty} \sin^2(\sqrt{n^2+n} \pi - n\pi) \\ &= \lim_{n \rightarrow \infty} \sin^2 \pi \left(\frac{n}{\sqrt{n^2+n} + n} \right) \end{aligned}$$

3. 正负

$$\textcircled{1} \lim_{n \rightarrow \infty} 2^{\infty} = \lim_{n \rightarrow \infty} 2^{+\infty} = \infty$$

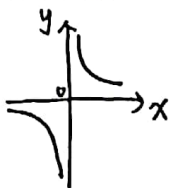
当 $\lim$ 下方为 n 时, ∞ 仅表示趋于 $+\infty$ 时的情况

$$\begin{aligned} \textcircled{2} i \lim_{x \rightarrow \infty} 2^x &< \lim_{x \rightarrow \infty} 2^{+\infty} = \infty \\ \lim_{x \rightarrow \infty} 2^{-\infty} &= \lim_{x \rightarrow -\infty} \frac{1}{2^{+\infty}} = \lim_{x \rightarrow \infty} \frac{1}{\infty} = 0 \end{aligned}$$





$$\text{ii } \lim_{x \rightarrow 0} \frac{1}{x} = \begin{cases} \lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty \\ \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty \end{cases}$$



等价无穷小代换, $(x \rightarrow 0)$

$$\text{① } \sqrt[n]{1+x} - 1 \sim \frac{1}{n}x$$

$$\text{i } \lim_{x \rightarrow 1} \frac{(\sqrt{x}-1)(\sqrt[3]{x}-1)(\sqrt[4]{x}-1) \cdots (\sqrt[n]{x}-1)}{(x-1)^{n-1}}$$

▲ 配凑

$$\text{原式} = \lim_{x \rightarrow 1} \frac{(\sqrt{1+(x-1)}-1)(\sqrt[3]{1+(x-1)}-1) \cdots (\sqrt[n]{1+(x-1)}-1)}{(x-1)^{n-1}}$$

$$= \lim_{x \rightarrow 1} \frac{\frac{1}{2}(x-1) \cdot \frac{1}{3}(x-1) \cdots \frac{1}{n}(x-1)}{(x-1)^{n-1}}$$

$$= \lim_{x \rightarrow 1} \frac{1}{n!}$$

$$\text{② ii } \lim_{x \rightarrow +\infty} (\sqrt[6]{x^6+x^5} - \sqrt[6]{x^6-x^5}) \quad \triangle \text{ 无穷大} \rightarrow \text{有限数}$$

$$\text{原式} = \lim_{x \rightarrow +\infty} x \left(\sqrt[6]{1+\frac{1}{x}} - \sqrt[6]{1-\frac{1}{x}} \right)$$

$$= \lim_{x \rightarrow +\infty} x \left[\sqrt[6]{1+\frac{1}{x}} - \left(\sqrt[6]{1+\left(-\frac{1}{x}\right)} - 1 \right) \right]$$

$$= \lim_{x \rightarrow +\infty} x \left(\frac{1}{6} \frac{1}{x} + \frac{1}{6} \frac{1}{x} \right)$$

$$= \frac{1}{3}$$

$$\text{② } e^x - 1 \sim x \quad (x \rightarrow 0)$$

$$\text{i } \lim_{x \rightarrow 0} \frac{e^{\sin x} - e^x}{\sin x - x}$$

$$\rightarrow \lim_{x \rightarrow 0} \frac{e^{\sin x} - 1 - (e^x - 1)}{\sin x - x}$$

$$= \lim_{x \rightarrow 0} \frac{e^x (e^{\sin x - x} - 1)}{\sin x - x}$$

$$= \lim_{x \rightarrow 0} e^x$$

$$= 1$$

$$\text{例: } \lim_{x \rightarrow 0} \left(\frac{2+e^{\frac{1}{x}}}{1+e^{\frac{1}{x}}} + \frac{\sin x}{|x|} \right)$$

$$\text{① 原式} = \lim_{x \rightarrow 0^-} \left(\frac{2+e^{\frac{1}{x}}}{1+e^{\frac{1}{x}}} - \frac{\sin x}{x} \right)$$

$$\frac{e}{x} \cdot \frac{x}{x} \rightarrow -\infty$$

$$= \lim_{x \rightarrow 0^-} \frac{2+e^{\frac{1}{x}}}{1+e^{\frac{1}{x}}} - \lim_{x \rightarrow 0^-} \frac{\sin x}{x}$$

$$= 2-1$$

$$= 1$$

$$\text{② } \lim_{x \rightarrow 0^+} \left(\frac{2+e^{\frac{1}{x}}}{1+e^{\frac{1}{x}}} + \frac{\sin x}{x} \right)$$

提取最大项

$$\lim_{x \rightarrow 0^+} \left(\frac{e^{\frac{1}{x}}(2e^{-\frac{1}{x}}+1)}{e^{\frac{1}{x}}(e^{-\frac{1}{x}}+1)} + \frac{\sin x}{x} \right)$$

$$= 0+1$$

$$= 1$$

常规极限 $\lim_{x \rightarrow a} f(x) = ?$ 的计算

$x=a$ $\left\{ \begin{array}{l} \text{分母不为0, 直接代入} \\ \text{分母为0, 分子不为零, 约去}(x-a) \\ \text{分子分母均为0, 无穷小代换} \\ \text{找不到可约分的因子, 洛必达} \end{array} \right.$

▲: 加减法不能用等价无穷小代换





$$\begin{aligned} & \lim_{n \rightarrow \infty} n^2 (2^{\frac{1}{n}} - 2^{\frac{1}{n+1}}) \\ &= \lim_{n \rightarrow \infty} n^2 2^{\frac{1}{n+1}} (2^{\frac{1}{n} - \frac{1}{n+1}} - 1) \\ &= \lim_{n \rightarrow \infty} 2^{\frac{1}{n+1}} \lim_{n \rightarrow \infty} n^2 (\frac{1}{n} - \frac{1}{n+1}) \ln 2 \\ &= \lim_{n \rightarrow \infty} \ln 2 \lim_{n \rightarrow \infty} 2^{\frac{1}{n+1}} \lim_{n \rightarrow \infty} \frac{n}{n+1} \\ &= \ln 2 \end{aligned}$$

③ 对数 $\ln(1+x) \sim x \quad x \rightarrow 0$

$$\begin{aligned} \text{i. } & \lim_{x \rightarrow 0} \frac{\ln \sin x - \ln x}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{\ln(\frac{\sin x}{x}) + 1 - 1}{x^2} \end{aligned}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\sin x}{x} - 1}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x - x}{x^3} \quad \boxed{x - \sin x \sim \frac{1}{6}x^3}$$

$$= \lim_{x \rightarrow 0} \frac{-\frac{1}{6}x^3}{x^3} \quad \text{泰勒展开}$$

$$= -\frac{1}{6}$$

$$\text{ii. } \lim_{x \rightarrow 0} \frac{\ln(\sin^2 x + e^x) - x}{\ln(x^2 + e^{2x}) - 2x}$$

$$= \lim_{x \rightarrow 0} \frac{\ln(\sin^2 x + e^x) - \ln e^x}{\ln(x^2 + e^{2x}) - \ln e^{2x}}$$

$$= \lim_{x \rightarrow 0} \frac{\ln \frac{\sin^2 x + e^x}{e^x}}{\ln \frac{x^2 + e^{2x}}{e^{2x}}}$$

$$= \lim_{x \rightarrow 0} \frac{\ln(1 + \frac{\sin^2 x}{e^x})}{\ln(1 + \frac{x^2}{e^{2x}})}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\sin^2 x}{e^x}}{\frac{x^2}{e^{2x}}} = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x}\right)^2 \lim_{x \rightarrow 0} e^x = 1$$

$$\text{iii. } \lim_{x \rightarrow 0} \frac{\ln(e^{\sin x} + \sqrt[3]{1 - \cos x}) - \sin x}{\arctan(4\sqrt[3]{1 - \cos x})}$$

$$\text{原式} = \lim_{x \rightarrow 0} \frac{\ln \frac{e^{\sin x} + \sqrt[3]{1 - \cos x}}{e^{\sin x}}}{\arctan(4\sqrt[3]{1 - \cos x})}$$

$$= \lim_{x \rightarrow 0} \frac{\ln(1 + \frac{\sqrt[3]{1 - \cos x}}{e^{\sin x}})}{\arctan(4\sqrt[3]{1 - \cos x})}$$

$$= \lim_{x \rightarrow 0} \frac{\sqrt[3]{1 - \cos x}}{e^{\sin x} \arctan(4\sqrt[3]{1 - \cos x})}$$

$$= \lim_{x \rightarrow 0} \frac{\sqrt[3]{1 - \cos x}}{4e^{\sin x} \sqrt[3]{1 - \cos x}}$$

$$= \frac{1}{4}$$

含有参数的函数

$$\lim_{x \rightarrow -1} \frac{x^3 - ax^2 - x + 4}{x+1} = l \quad \text{求 } a, l \text{ 的值}$$

▲: 分母极限为0, 整体要想存在极限, 则分子极限为0

$$\text{i. } \lim_{x \rightarrow -1} x^3 - ax^2 - x + 4 = 0 \quad -1 - a + 1 + 4 = 0 \quad \therefore a = 4$$

$$\lim_{x \rightarrow -1} \frac{x^3 - 4x^2 - x + 4}{x+1} = l$$

$$\lim_{x \rightarrow -1} \frac{(x-4)(x+1)(x+1)}{x+1} = l \quad \lim_{x \rightarrow -1} (x-4)(x+1) = l$$

$$\therefore l = 10$$

$$\text{ii. } \lim_{x \rightarrow 2} \frac{x^2 + ax + b}{x^2 - x - 2} = 2$$

$$\lim_{x \rightarrow 2} \frac{x^2 + ax + b}{(x-2)(x+1)} = 2$$

$$\text{原式} = \lim_{x \rightarrow 2} \frac{x+c}{x+1}$$

$$= \frac{2+c}{3}$$

$$\therefore \frac{2+c}{3} = 2 \quad c = 4$$

$$\star \text{ 设 } x^2 + ax + b = (x-2)(x+c)$$

$$a = c - 2 \quad b = -2c$$

$$\therefore a = 2 \quad b = -8$$





例: $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^p} = 1$ 求 p 的值。

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{2} x^3}{x^p} \quad p=3 \quad 1=\frac{1}{2}$$

例: $\lim_{x \rightarrow 0} \frac{1 + \frac{x^2}{2} - \sqrt{1+x^2}}{(\cos x - e^{x^2}) \sin(x^3)}$

$$= \lim_{x \rightarrow 0} \frac{(1 + \frac{x^2}{2})^2 - (1+x^2)}{x^2(\cos x - e^{x^2})} \cdot \frac{1}{1 + \frac{x^2}{2} + \sqrt{1+x^2}}$$

$$= \lim_{x \rightarrow 0} \frac{x^2}{4(\cos x - e^{x^2})} \lim_{x \rightarrow 0} \frac{1}{1 + \frac{x^2}{2} + \sqrt{1+x^2}}$$

$$= \frac{1}{8} \lim_{x \rightarrow 0} \frac{\cos x - 1 - (e^{x^2} - 1)}{x^2}$$

$$= \frac{1}{8} \lim_{x \rightarrow 0} \frac{\cos x - 1 - e^{x^2} + 1}{x^2}$$

$$= \frac{1}{8} \lim_{x \rightarrow 0} \frac{\cos x - 1}{x^2} - \lim_{x \rightarrow 0} \frac{e^{x^2} - 1}{x^2} \quad \star$$

$$= \frac{1}{8} \cdot \frac{1}{-\frac{1}{2} - 1}$$

$$= -\frac{1}{12}$$

转化与联想

例: 已知极限 $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} = 0$

求 $\lim_{x \rightarrow 0} \frac{1-f(x)}{x^2}$ 的值

$$\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} + \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \lim_{x \rightarrow 0} \frac{1-f(x)}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{2} x^3}{x^3} + \lim_{x \rightarrow 0} \frac{\sin x}{x} \lim_{x \rightarrow 0} \frac{1-f(x)}{x^2}$$

$$= \frac{1}{2} + \lim_{x \rightarrow 0} \frac{1-f(x)}{x^2} = 0$$

$$\therefore \lim_{x \rightarrow 0} \frac{1-f(x)}{x^2} = -\frac{1}{2}$$

例: 已知 $\lim_{x \rightarrow 1} f(x)$ 存在且 $f(x) = 2x^2 + 5 + 3 \lim_{x \rightarrow 1} f(x)$

求 $f(x)$

$\because \lim_{x \rightarrow 1} f(x)$ 存在, 不妨 $\lim_{x \rightarrow 1} f(x) = A$

$$f(x) = 2x^2 + 5 + 3A \quad \lim_{x \rightarrow 1} f(x) = 7 + 3A = A$$

$$\text{得 } A = -\frac{7}{2} \quad f(x) = 2x^2 + 5 - \frac{21}{2} = 2x^2 - \frac{11}{2}$$

练: 已知 $\lim_{x \rightarrow \infty} f(x)$ 存在且

$\lim_{x \rightarrow \infty} 3xf(x) = \lim_{x \rightarrow \infty} [4f(x) + 5]$ 求 $\lim_{x \rightarrow \infty} xf(x)$ 的值

解: 设 $\lim_{x \rightarrow \infty} f(x) = A$ $\lim_{x \rightarrow \infty} 3xf(x) = 3A + 5$

$$3 \lim_{x \rightarrow \infty} xf(x) = 3A + 5 \quad \lim_{x \rightarrow \infty} xf(x) = A + \frac{5}{3}$$

$\star \because x \rightarrow \infty$ 时 $x \rightarrow \infty \quad f(x) \rightarrow 0$ 时 $\lim_{x \rightarrow \infty} xf(x)$ 存在

$$\therefore \lim_{x \rightarrow \infty} f(x) = 0 \quad A = 0 \quad \therefore \lim_{x \rightarrow \infty} xf(x) = \frac{5}{3}$$

无穷项运算转化为有限项运算。

Δ : 四则运算不满足无穷项问题。

$$\lim_{n \rightarrow \infty} \frac{k}{n^2} = 0 \quad \lim_{n \rightarrow \infty} \left(\frac{1}{n^2} + \frac{2}{n^2} + \dots + \frac{k}{n^2} + \dots + \frac{n}{n^2} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1+n}{n^2}$$

$$= \frac{1}{2} \neq \lim_{n \rightarrow \infty} \frac{1}{n^2} + \lim_{n \rightarrow \infty} \frac{2}{n^2} + \dots + \lim_{n \rightarrow \infty} \frac{n}{n^2} = 0$$

$$\neq \lim_{n \rightarrow \infty} \left(\frac{1}{n^2} + \frac{1}{n^2} + \dots + \frac{1}{n^2} \right)$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e \neq \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right) \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right) \dots \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right) = 1$$





初等法: 无穷级数为有限项

例: $\lim_{n \rightarrow \infty} (1 + x + x^2 + \dots + x^n)$, 其中 $|x| < 1$

等比数列求和公式:

$$\text{原式} = \lim_{n \rightarrow \infty} \frac{1 - x^{n+1}}{1 - x}$$

$$\because |x| < 1 \therefore n \rightarrow \infty \text{ 时 } x^{n+1} \rightarrow 0$$

$$\therefore \text{原式} = \frac{1}{1-x}$$

例: $\lim_{n \rightarrow \infty} \left[\frac{1}{2} + \frac{1}{6} + \dots + \frac{1}{n(n+1)} \right]$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{n(n+1)} \right]$$

$$= \lim_{n \rightarrow \infty} \left(1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots + \frac{1}{n} - \frac{1}{n+1} \right)$$

$$= 1$$

$$* \frac{1}{k(k+1)(k+2)} = \left[\frac{1}{2k} - \frac{1}{2(k+1)} \right] - \left[\frac{1}{2(k+1)} - \frac{1}{2(k+2)} \right]$$

$$\lim_{n \rightarrow \infty} \left[\frac{1}{6} + \frac{1}{24} + \dots + \frac{1}{n(n+1)(n+2)} \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{1 \times 2 \times 3} + \frac{1}{2 \times 3 \times 4} + \dots + \frac{1}{n(n+1)(n+2)} \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{2} - \frac{1}{4} - \frac{1}{4} + \frac{1}{6} + \frac{1}{4} - \frac{1}{6} - \frac{1}{6} + \frac{1}{8} + \dots + \frac{1}{2n} - \frac{1}{n+1} - \frac{1}{2(n+2)} \right]$$

$$= \lim_{n \rightarrow \infty} \left[\left(\frac{1}{2} - \frac{1}{4} \right) - \left(\frac{1}{4} - \frac{1}{6} \right) + \left(\frac{1}{4} - \frac{1}{6} \right) - \left(\frac{1}{6} - \frac{1}{8} \right) + \dots + \left(\frac{1}{2n} - \frac{1}{2(n+1)} \right) - \left(\frac{1}{2(n+1)} - \frac{1}{2(n+2)} \right) \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{2} - \frac{1}{4} - \frac{1}{2(n+1)} + \frac{1}{2(n+2)} \right]$$

$$= \frac{1}{4}$$

例: $\lim_{n \rightarrow \infty} \frac{1}{2}$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{2^2} \right) \left(1 - \frac{1}{3^2} \right) \dots \left(1 - \frac{1}{n^2} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{2^2-1}{2^2} \times \frac{3^2-1}{3^2} \times \dots \times \frac{n^2-1}{n^2}$$

$$= \lim_{n \rightarrow \infty} \frac{(2+1)(2-1)}{2^2} \times \frac{(3+1)(3-1)}{3^2} \times \dots \times \frac{(n+1)(n-1)}{n^2}$$

$$= \lim_{n \rightarrow \infty} \prod_{k=2}^n \frac{(k-1)(k+1)}{k^2}$$



$$= \lim_{n \rightarrow \infty} \prod_{k=2}^n \frac{k-1}{k} \times \prod_{k=2}^n \frac{k+1}{k}$$

$$\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \dots \times \frac{n-1}{n} \times \frac{3}{2} \times \frac{4}{3} \times \dots \times \frac{n}{n-1} \times \frac{n+1}{n}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \times \frac{n+1}{2}$$

$$= \frac{1}{2}$$

[变]

例: $\lim_{n \rightarrow \infty} \left(1 - \frac{2}{2^3+1} \right) \left(1 - \frac{2}{3^3+1} \right) \dots \left(1 - \frac{2}{n^3+1} \right)$

$$= \lim_{n \rightarrow \infty} \prod_{k=2}^n \frac{k^3-1}{k^3+1}$$

$$= \lim_{n \rightarrow \infty} \prod_{k=2}^n \frac{(k-1)(k^2+k+1)}{(k+1)(k^2-k+1)}$$

$$= \lim_{n \rightarrow \infty} \prod_{k=2}^n \frac{(k-1)}{(k+1)} \prod_{k=2}^n \frac{(k^2+k+1)}{(k^2-k+1)}$$

$$\frac{1}{3} \times \frac{2}{4} \times \frac{3}{5} \times \dots \times \frac{n-1}{n} \times \frac{n}{n+1}$$

$$\frac{(k+1)^2 - (k+1) + 1}{k^2 - k + 1} \quad \text{写成相似的形式 (k+1)/(k+1)}$$

$$\therefore \text{原式} = \lim_{n \rightarrow \infty} \frac{2}{n(n+1)} \frac{(n+1)^2 - (n+1) + 1}{2^2 - 2 + 1}$$

$$= \lim_{n \rightarrow \infty} \frac{2}{n(n+1)} \frac{n^2 + n + 1}{3}$$

$$= \frac{2}{3}$$





例: $\lim_{n \rightarrow \infty} (1+x)(1+x^2)(1+x^4) \cdots (1+x^{2^n})$, 其中 $|x| < 1$

$$\begin{aligned} \text{原式} &= \lim_{n \rightarrow \infty} \frac{(1-x)(1+x)(1+x^2)(1+x^4) \cdots (1+x^{2^n})}{1-x} \\ &= \lim_{n \rightarrow \infty} \frac{1-x^{2^{n+1}}}{1-x} \\ &= \frac{1}{1-x} \end{aligned}$$

例: $\lim_{n \rightarrow \infty} \cos \frac{x}{2} \cos \frac{x}{4} \cos \frac{x}{8} \cdots \cos \frac{x}{2^n}$

$$\text{原式} = \lim_{n \rightarrow \infty} \frac{\cos \frac{x}{2} \cos \frac{x}{4} \cos \frac{x}{8} \cdots \cos \frac{x}{2^n} \cdot \sin \frac{x}{2^n}}{\sin \frac{x}{2^n}}$$

$$= \lim_{n \rightarrow \infty} \frac{\frac{1}{2^n} \sin x}{\sin \frac{x}{2^n}} \quad \text{注意变量为 } n$$

$$= \lim_{n \rightarrow \infty} \frac{\frac{1}{2^n} \sin x}{\frac{x}{2^n}} \quad \text{当 } n \rightarrow \infty \text{ 时 } \frac{x}{2^n} \rightarrow 0$$

$$= \frac{\sin x}{x}$$

无穷项变为有限项 — 夹逼准则

例: $\lim_{n \rightarrow \infty} (\frac{1}{n^2+1} + \frac{1}{n^2+2} + \cdots + \frac{1}{n^2+n})$

$$\frac{\frac{1}{n^2+n}}{\frac{1}{n^2+n}} < (\quad) < \frac{\frac{1}{n^2+n}}{\frac{1}{n^2+n}} \quad \text{最小分母}$$

▲: 分母近似相等时, 可以利用夹逼定理
化为同分母 (充分性条件)

$$\text{例: } \frac{n}{(n+1)^3} \leq \frac{1}{(n+1)^3} + \frac{1}{(n+2)^3} + \cdots + \frac{1}{(n+n)^3} \leq \frac{n}{(n+1)^3}$$

$$\therefore \lim_{n \rightarrow \infty} \left[\frac{1}{(n+1)^3} + \frac{1}{(n+2)^3} + \cdots + \frac{1}{(n+n)^3} \right] = 0$$

例: $\lim_{n \rightarrow \infty} \sqrt[n]{1+2^n+3^n}$

$$= \lim_{n \rightarrow \infty} \sqrt[n]{3^n \left(\frac{1}{3^n} + \frac{2^n}{3^n} + 1 \right)}$$

$$= \lim_{n \rightarrow \infty} 3 \sqrt[n]{\frac{1}{3^n} + \frac{2^n}{3^n} + 1}$$

$$= 3$$

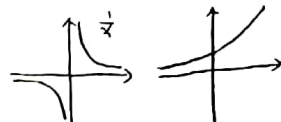
1. 设 $x \rightarrow 0$ 时, $(1-\cos x) \ln(1+x^2)$ 是比 $x \sin x^n$ 高阶的无穷小, 而 $x \sin x^n$ 是比 $e^{x^2} - 1$ 高阶的无穷小, 则正整数 $n = 2$

$$\lim_{x \rightarrow 0} \frac{(1-\cos x) \ln(1+x^2)}{x \sin x^n} = \lim_{x \rightarrow 0} \frac{\frac{1}{2} x^2 \cdot x^2}{x \cdot x^n} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{x^4}{x^{n+1}}$$

$$n+1 < 4 \quad n < 3$$

$$\lim_{x \rightarrow 0} \frac{x \sin x^n}{e^{x^2} - 1} = \lim_{x \rightarrow 0} \frac{x^{n+1}}{x^2} \quad n+1 > 2 \quad n > 1$$

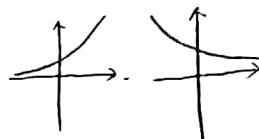
$$2. \text{求极限: } \lim_{x \rightarrow 0} \frac{1}{2+3^{\frac{1}{x}}}$$



$$\text{① } \lim_{x \rightarrow 0^+} \frac{1}{2+3^{\frac{1}{x}}} = 0$$

$$(\lim_{x \rightarrow 0^+} 2+3^{\frac{1}{x}} = \infty)$$

$$\text{② } \lim_{x \rightarrow 0^-} \frac{1}{2+3^{\frac{1}{x}}} = \frac{1}{2}$$



3. 求极限 $\lim_{n \rightarrow \infty} \frac{a^n}{1+a^n} \quad (a > 0)$

$$\lim_{n \rightarrow \infty} \frac{1+a^n-1}{1+a^n} = \lim_{n \rightarrow \infty} 1 - \frac{1}{1+a^n} \quad \text{若 } |a| > 1 \quad 1+a^n \rightarrow \infty$$

$$\text{原式} = \lim_{n \rightarrow \infty} 1 - \frac{1}{1+a^n} = 1 \quad \text{若 } |a| < 1 \quad \text{原式} = 0$$

4. $\lim_{n \rightarrow \infty} \frac{\ln(a^n + e^n)}{n} \quad (a > 0)$

$$\text{原式} = \lim_{n \rightarrow \infty} \frac{\ln e^n \left[\left(\frac{a}{e} \right)^n + 1 \right]}{n}$$

$$= \lim_{n \rightarrow \infty} \frac{n + \ln \left[\left(\frac{a}{e} \right)^n + 1 \right]}{n}$$

$$= \lim_{n \rightarrow \infty} 1 + \frac{1}{n} \ln \left[\left(\frac{a}{e} \right)^n + 1 \right]$$

$$\text{① } 0 < a \leq e \quad \text{原式} = 1$$

$$\text{② } a > e \quad \lim_{n \rightarrow \infty} \frac{\ln a^n \left[1 + \left(\frac{e}{a} \right)^n \right]}{n}$$

$$= \lim_{n \rightarrow \infty} \frac{n \ln a + \ln \left[1 + \left(\frac{e}{a} \right)^n \right]}{n}$$

$$= \ln a$$





讨论 $\lim_{n \rightarrow \infty} \frac{\ln(a^n + e^n)}{n}$ ($a > 0$) 的连续性

$$f(a) = \begin{cases} 1 & 0 < a \leq e \\ \ln a & a > e \end{cases}$$

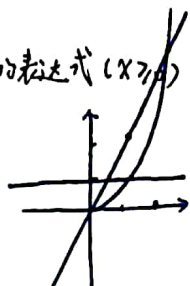
$$f(e^-) = f(e) = f(e^+) = 1$$

$\therefore f(x)$ 在 $x=0$ 时处处连续

例: 求 $f(x) = \lim_{n \rightarrow \infty} \sqrt[n]{1 + (2x)^n + x^{2n}}$ 的表达式 ($x \geq 0$)

并讨论其连续性

$$f(x) = \lim_{n \rightarrow \infty} \sqrt[n]{1 + (2x)^n + (x^2)^n}$$



*: $0 \leq x < \frac{1}{2}$ 时, $1, 2x, x^2$ 中 1 最大
 $\frac{1}{2} \leq x < 2$ 时 $2x$ 最大
 $x \geq 2$ 时, x^2 最大.

① $0 \leq x < \frac{1}{2}$ 时

$$\sqrt[n]{1} \leq \sqrt[n]{1 + (2x)^n + (x^2)^n} \leq 1 \cdot \sqrt[n]{3}$$

② $\frac{1}{2} \leq x < 2$ 时有

$$\sqrt[n]{(2x)^n} \leq \sqrt[n]{1 + (2x)^n + (x^2)^n} \leq 2x \sqrt[n]{3}$$

③ $x \geq 2$ 时有

$$\sqrt[n]{(x^2)^n} \leq \sqrt[n]{1 + (2x)^n + (x^2)^n} \leq x^2 \sqrt[n]{3}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{3} = \lim_{n \rightarrow \infty} 3^{\frac{1}{n}} = 1$$

$$\therefore f(x) = \begin{cases} 1 & 0 \leq x < \frac{1}{2} \\ 2x & \frac{1}{2} \leq x < 2 \\ x^2 & x \geq 2 \end{cases}$$

$$\lim_{x \rightarrow \frac{1}{2}^-} f(x) = 1 = \lim_{x \rightarrow \frac{1}{2}^+} f(x) = f\left(\frac{1}{2}\right)$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2) = 4$$

$\therefore f(x)$ 在 $[0, +\infty)$ 上处处连续

练: 讨论函数 $f(x) = \lim_{n \rightarrow \infty} \sqrt[n]{1 + x^n + \left(\frac{x^2}{2}\right)^n}$ 的连续性

① $0 < x < 1$

$$1 \leq \sqrt[n]{1 + x^n + \left(\frac{x^2}{2}\right)^n} \leq 1 \cdot \sqrt[n]{3}$$

② $1 \leq x < 2$

$$x \leq \sqrt[n]{1 + x^n + \left(\frac{x^2}{2}\right)^n} \leq x \sqrt[n]{3}$$

③ $x \geq 2$

$$\frac{x^2}{2} \leq \sqrt[n]{1 + x^n + \left(\frac{x^2}{2}\right)^n} \leq \frac{x^2}{2} \sqrt[n]{3}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{3} = 1 \quad \text{夹逼定理}$$

$$f(x) = \begin{cases} 1 & 0 < x < 1 \\ x & 1 \leq x < 2 \\ \frac{x^2}{2} & x \geq 2 \end{cases}$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1) = 1$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2) = 2$$

$\therefore f(x)$ 在 $[0, +\infty)$ 上处处连续.

单调有界原理

定理: 单调有界数列必有极限.

先证单调性/有界性依题而定

常见类型

1) 先单调后有界

2) 先有界后单调

3) 先证有上界, 再证 $\nearrow$

或先证有下界, 再证 $\searrow$

4) 通过相邻项之差的同号性, 证明单调性

